Negative electricity prices in Germany, 2024–2025

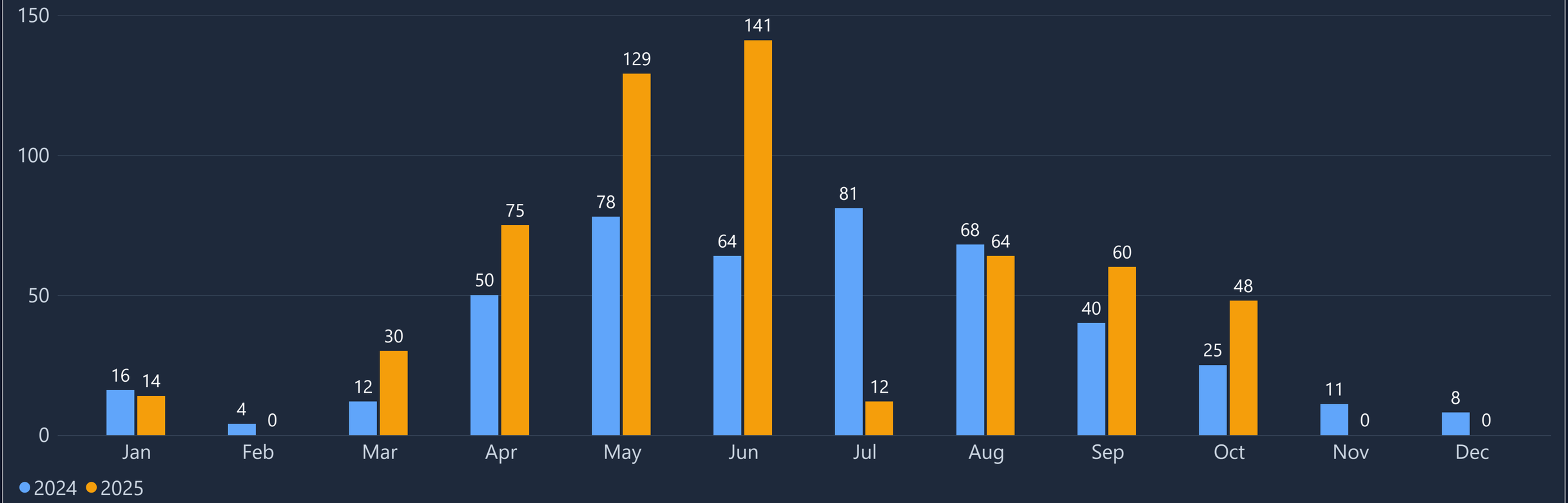
Select Year

2024 | 2025

Hours Analysed	Negative Price Hours	Share of All Hours with a Negative Price
17,544	1,030	5.87%

Negative-price hours peak in spring and summer, but July 2025 collapsed to just 12

Number of hours per month with a day-ahead price below zero, 2024 vs 2025

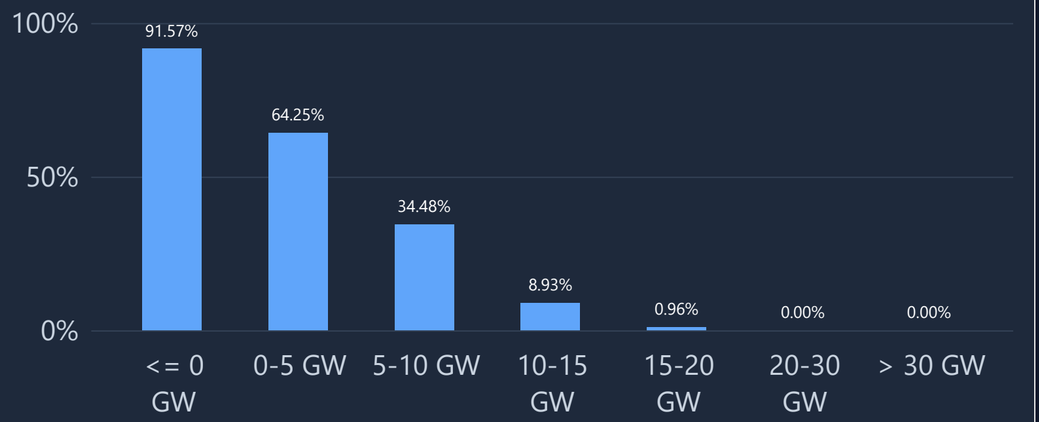


Source: Bundesnetzagentur | SMARD.de (CC BY 4.0). Day-ahead prices, hourly, Germany/Luxembourg bidding zone.

Negative prices cluster at midday, in spring and summer, and on weekends

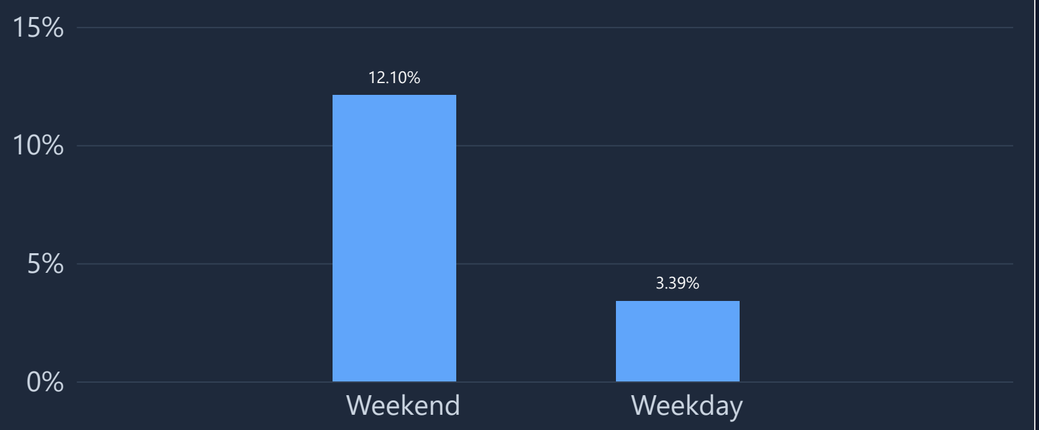
92% of hours at zero or negative residual load have a negative price

Residual load = grid demand minus wind and solar



Weekends are 3.6 times more likely to be negative than weekdays

12.1% of weekend hours vs 3.4% of weekday hours



Negative prices cluster around 12:00–15:00 in May and June

Share of hours with a negative price, by local hour (rows) and month (columns), 2024–2025

Hour	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	**Total**
0	1.61%	0.00%	0.00%	0.00%	0.00%	0.00%	1.61%	0.00%	3.33%	3.23%	1.67%	0.00%	**0.96%**
1	3.23%	1.75%	0.00%	0.00%	0.00%	0.00%	1.61%	1.61%	1.67%	6.45%	1.67%	1.61%	**1.64%**
2	3.23%	1.75%	0.00%	0.00%	0.00%	0.00%	1.61%	1.61%	3.33%	6.25%	1.67%	1.61%	**1.78%**
3	6.45%	1.75%	0.00%	0.00%	0.00%	0.00%	1.61%	1.61%	3.33%	6.45%	3.33%	1.61%	**2.19%**
4	6.45%	1.75%	0.00%	0.00%	0.00%	0.00%	1.61%	1.61%	3.33%	8.06%	3.33%	3.23%	**2.46%**
5	6.45%	0.00%	0.00%	0.00%	0.00%	0.00%	1.61%	1.61%	3.33%	4.84%	3.33%	1.61%	**1.92%**
6	3.23%	0.00%	0.00%	0.00%	0.00%	0.00%	1.61%	1.61%	0.00%	4.84%	1.67%	1.61%	**1.23%**
7	3.23%	0.00%	0.00%	0.00%	0.00%	0.00%	1.61%	1.61%	0.00%	4.84%	1.67%	1.61%	**1.23%**
8	1.61%	0.00%	1.61%	1.67%	0.00%	3.33%	1.61%	1.61%	0.00%	6.45%	0.00%	0.00%	**1.50%**
9	1.61%	0.00%	1.61%	3.33%	3.23%	13.33%	3.23%	3.23%	3.33%	6.45%	0.00%	0.00%	**3.28%**
10	1.61%	0.00%	3.23%	10.00%	19.35%	21.67%	4.84%	19.35%	6.67%	6.45%	0.00%	0.00%	**7.80%**
11	1.61%	0.00%	9.68%	23.33%	43.55%	36.67%	9.68%	20.97%	13.33%	6.45%	0.00%	0.00%	**13.82%**
12	1.61%	0.00%	17.74%	31.67%	53.23%	50.00%	19.35%	27.42%	21.67%	6.45%	0.00%	0.00%	**19.15%**
13	1.61%	0.00%	17.74%	38.33%	56.45%	55.00%	29.03%	35.48%	35.00%	11.29%	0.00%	0.00%	**23.39%**
14	1.61%	0.00%	8.06%	40.00%	58.06%	56.67%	25.81%	37.10%	35.00%	11.29%	0.00%	0.00%	**22.85%**
15	1.61%	0.00%	4.84%	35.00%	54.84%	48.33%	20.97%	30.65%	23.33%	6.45%	0.00%	0.00%	**18.88%**
16	1.61%	0.00%	1.61%	21.67%	35.48%	38.33%	12.90%	20.97%	8.33%	4.84%	0.00%	0.00%	**12.18%**
17	0.00%	0.00%	1.61%	3.33%	9.68%	13.33%	9.68%	4.84%	1.67%	1.61%	0.00%	0.00%	**3.83%**
18	0.00%	0.00%	0.00%	0.00%	0.00%	5.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	**0.41%**
19	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	**0.00%**
20	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	**0.00%**
21	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	1.61%	0.00%	0.00%	**0.14%**
22	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	1.61%	0.00%	0.00%	**0.14%**
23	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	1.61%	0.00%	0.00%	**0.14%**
Total	**2.02%**	**0.29%**	**2.83%**	**8.68%**	**13.91%**	**14.24%**	**6.25%**	**8.87%**	**6.94%**	**4.90%**	**0.76%**	**0.54%**	**5.87%**

Source: Bundesnetzagentur | SMARD.de (CC BY 4.0). Day-ahead prices, hourly, Germany/Luxembourg bidding zone.

Solar earns less than the average price, and the gap widened in 2025

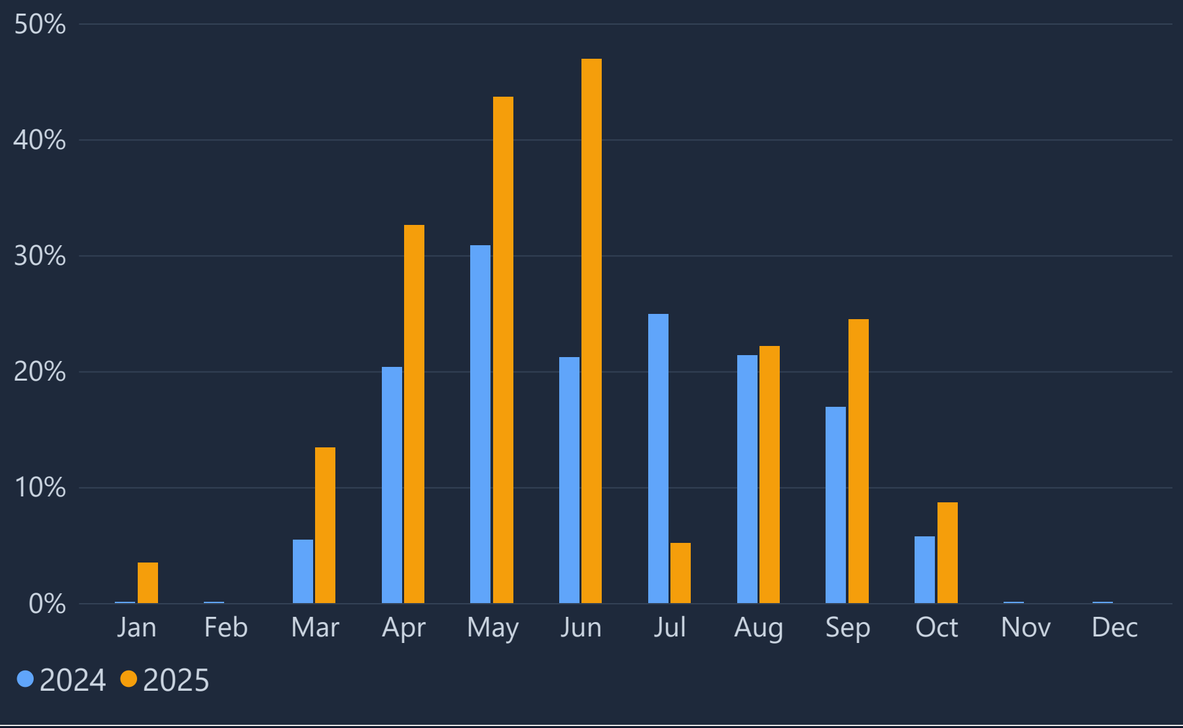
Year	Average price (€/MWh)	Solar-weighted price (€/MWh)	Capture rate	Solar output at negative prices
2024	78.5	46.2	0.589	18.4%
2025	89.3	46.1	0.516	24.1%
Total	**83.9**	**46.1**	**0.550**	**21.5%**

Solar's capture rate fell from 0.59 to 0.52, and hit 0.31 in June 2025

Capture rate = solar-weighted price ÷ average price. 1.0 means solar earns the average

More solar output sold at negative prices: 18% in 2024, 24% in 2025

Share of each month's solar generation produced in negative-price hours



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